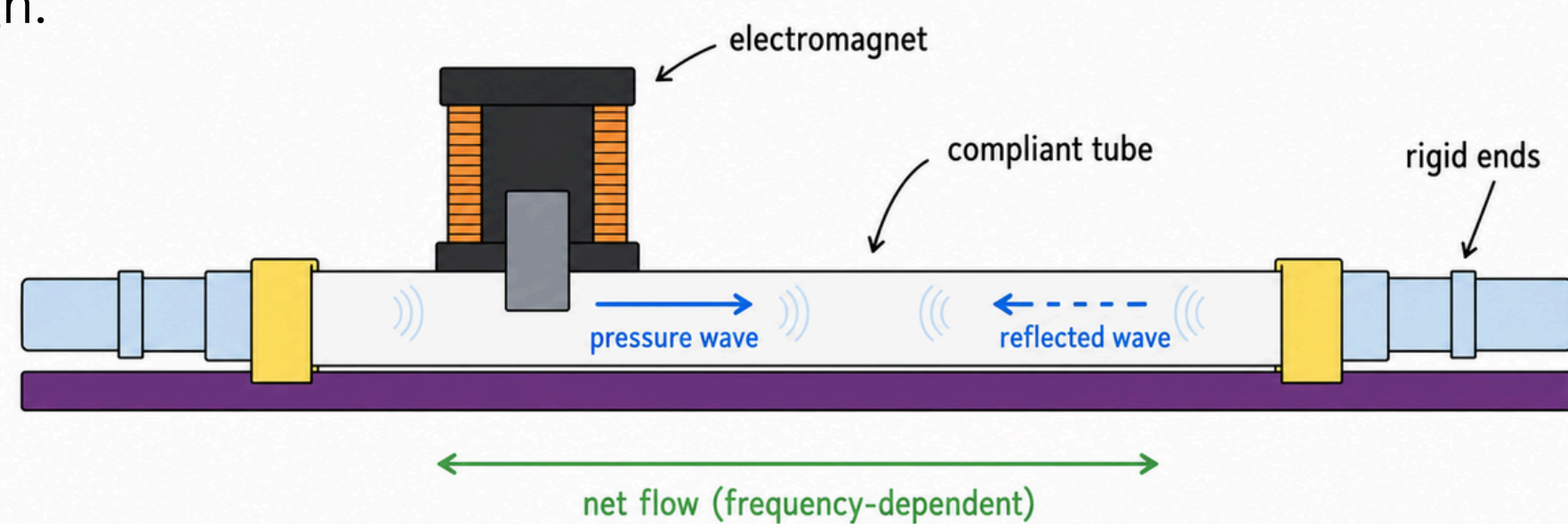


Motivation & Background

Liebau pumping, inspired by the mechanism of embryonic hearts, generates net fluid flow through compressions of a compliant tube with rigid ends [1]. Asymmetric compressions cause pressure waves to reflect off the tube's ends, creating an impedance mismatch. This generates flow that moves the robot forward or backward depending on the compression frequency. Our project aims to create an **energy-efficient** Liebau-powered underwater robot, with a focus on lowering power consumption through improved actuation design.



This work on improving the energy efficiency of the swimmer robot will help guide medical applications, particularly the Fontan procedure, an open-heart operation done on children born with only one functional ventricle rather than two [2]. Through design, optimization, and characterization of this robot, we aim to increase blood flow while addressing the challenges of power-efficient actuation for biomedical applications.

Electromagnet Tapping Actuator

To establish a baseline for comparison against alternative actuation strategies, we first optimized the existing electromagnet actuator. Flow experiments were conducted in an **open-loop system** rather than by directly measuring the robot's velocity and thrust.

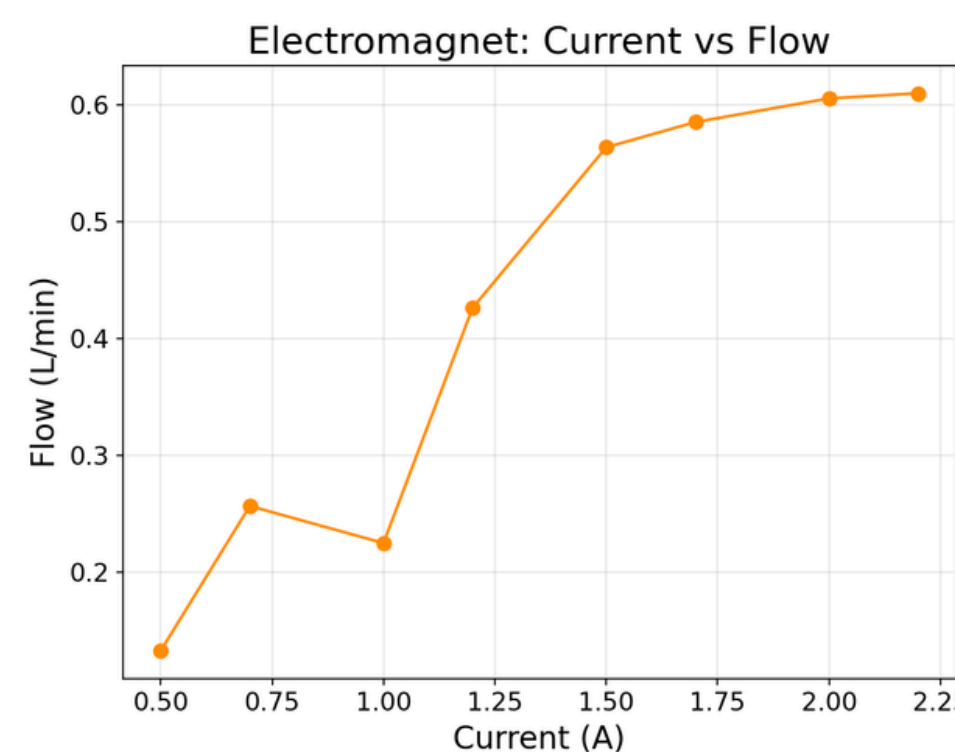
- Given **fixed constraints** (e.g., geometry of actuator, magnetization of plunger), we manipulated the following:
 - Current
 - Gauge of copper wire
 - Number of turns

Coil Optimization Formulas:

$$F = k_{\text{geo}} \cdot M \cdot V \cdot \left(NI \cdot \frac{dB}{dz} \right)$$

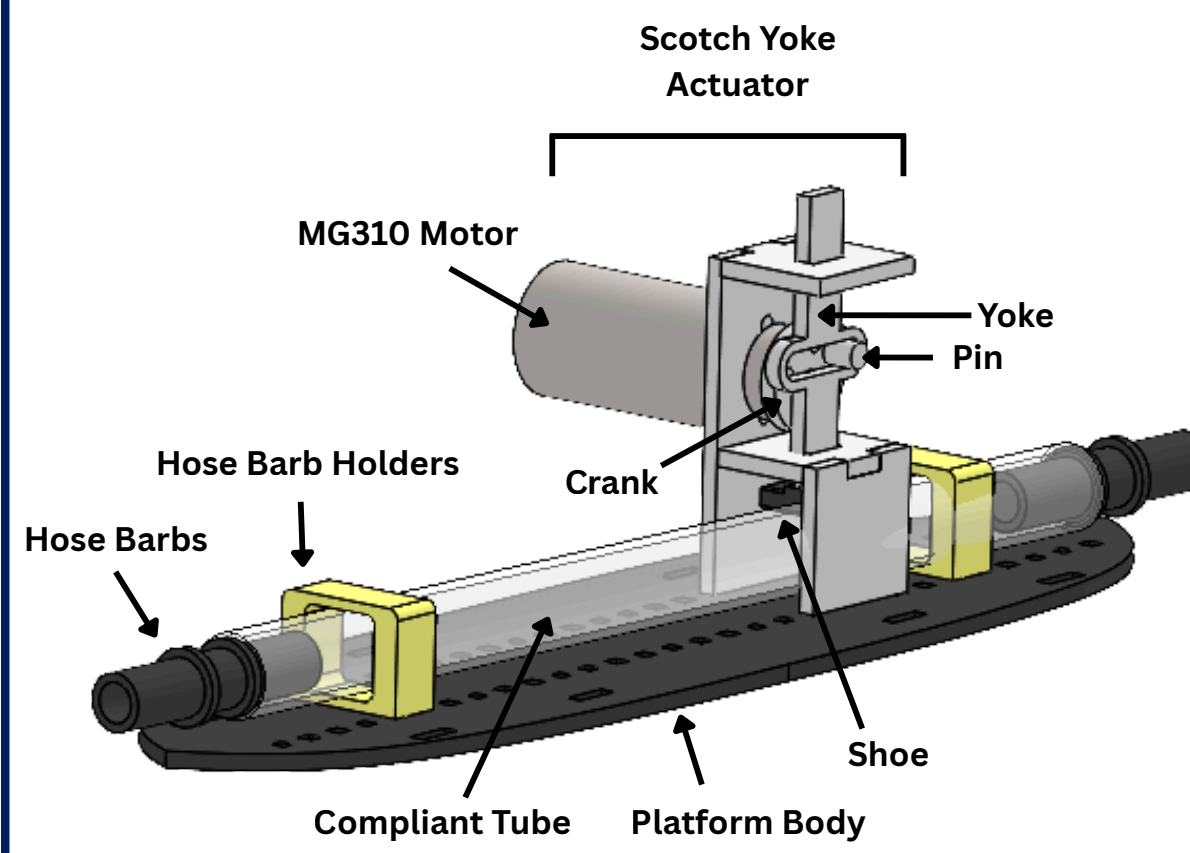
$$B_z(z) = \frac{\mu_0 NI}{2L} \left[\frac{z + \frac{L}{2}}{\sqrt{(z + \frac{L}{2})^2 + r_{\text{eff}}^2}} - \frac{z - \frac{L}{2}}{\sqrt{(z - \frac{L}{2})^2 + r_{\text{eff}}^2}} \right]$$

- This lowered the **robot's cost of transport**, halving theoretical power consumption at a constant magnetic plunger force.



Scotch Yoke Tapping Actuator

The **scotch yoke mechanism** converts rotational motion into linear motion, allowing a rotational motor to power the actuator's up/down pinching motion. We explore this alternative actuation method to lower power consumption in comparison to an electromagnet.



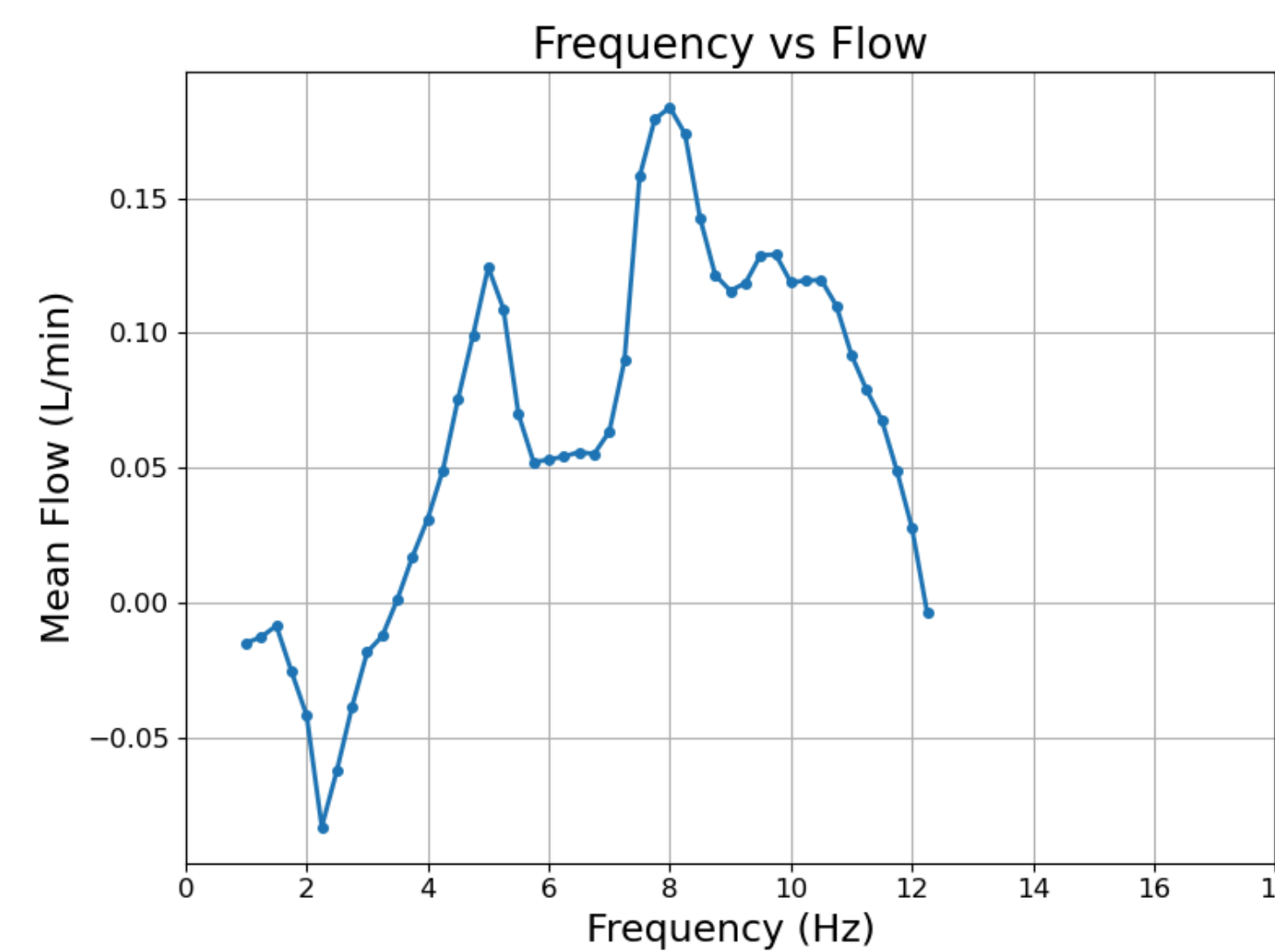
Components:

- Crank:** a disk rotating about its center, with a pin fixed at the edge. Crank diameter sets the stroke length
- Yoke:** a sliding element with a long slot that tracks the crank pin. The slot allows the pin to slide side-to-side while restricting yoke motion to up/down
- Motor:** supplies rotational power, spinning the crank
- Frame:** constrains the yoke to a single linear path, preventing drift

Parameters:

This motor-powered actuation method allows us to test different parameters affecting fluid flow.

- Occlusion**, the degree to which the actuator pinches the tube, affects the impedance mismatch that creates flow. It can be adjusted by changing the stroke length, which is set by the crank diameter.
- Contact area** between the shoe and the tube can be varied by swapping in shoe attachments of different sizes on the bottom end of the yoke.
- Impact force** is controlled by the velocity at the point of contact, adjusted by varying speed or dwell time at the bottom of the stroke.

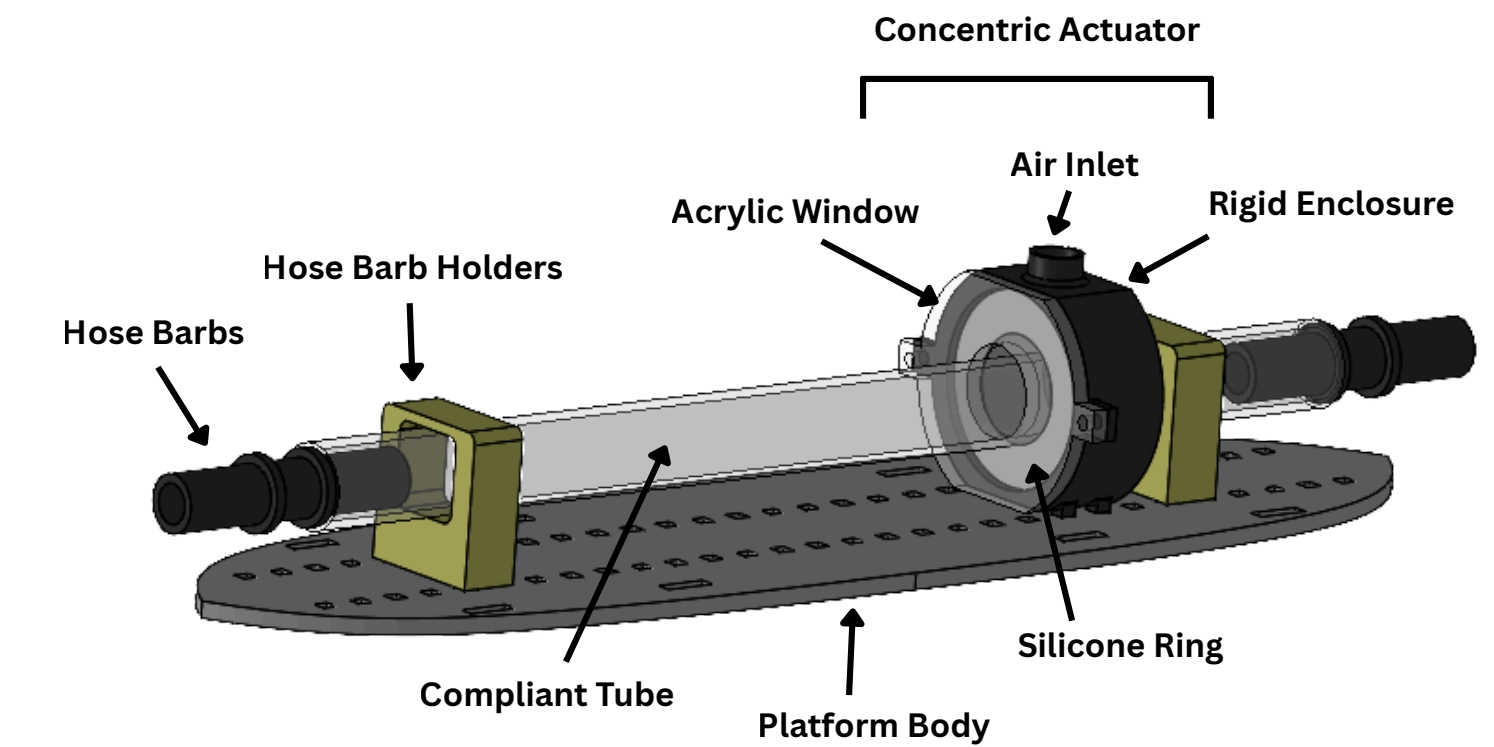


Concentric Pinching Actuator

Concentric pinching distributes pressure evenly across the tube's circumference rather than at a single point, making it suitable for medical applications.

Testing Objectives:

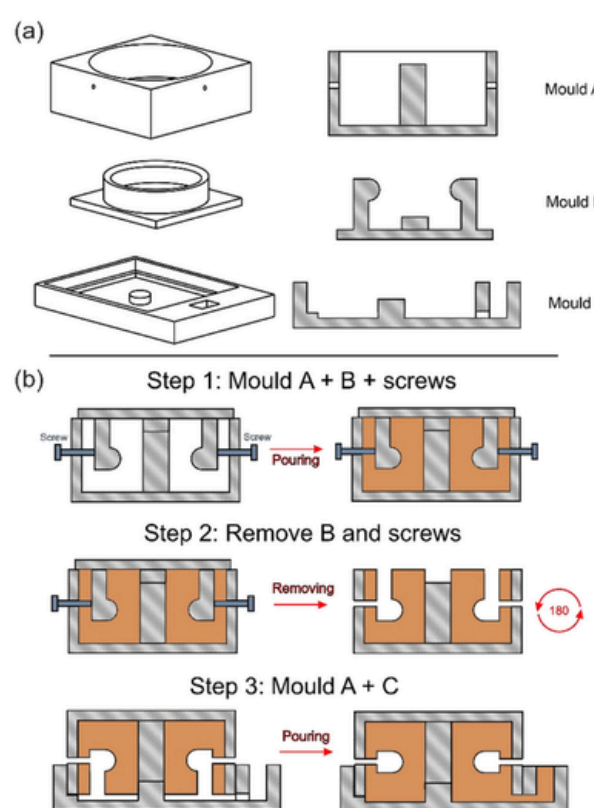
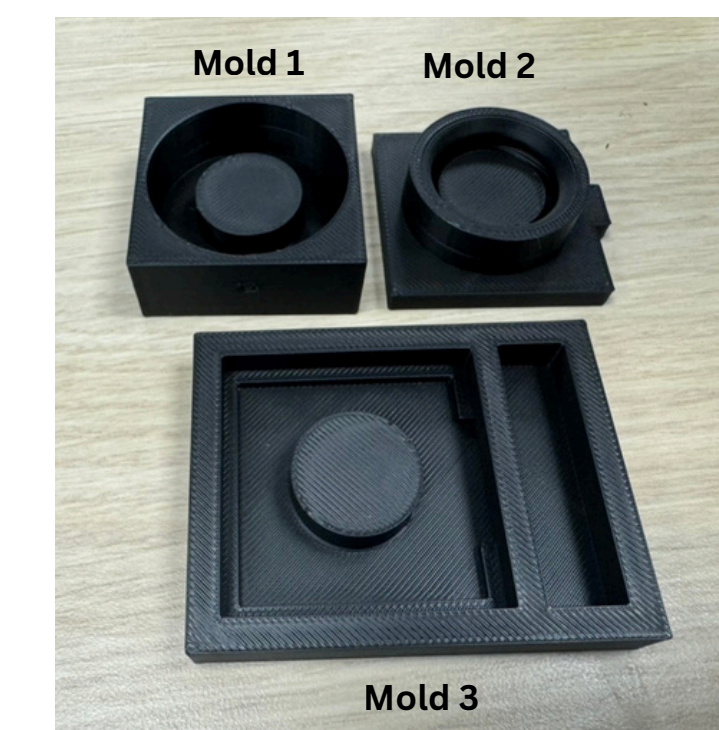
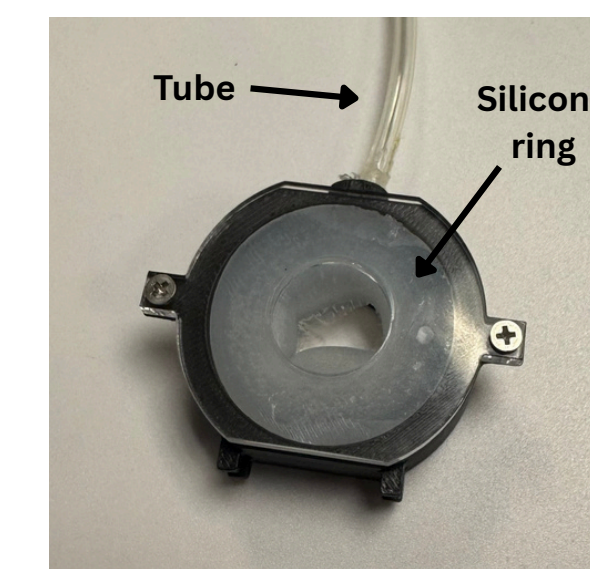
- Determine whether the concentric actuator is more energy-efficient than the tapping actuator
- Determine whether the concentric actuator can compress and decompress quickly enough at the target frequency of 11 Hz, which generates the most net fluid flow



Design:

The concentric actuator consists of a hollow silicone ring that wraps around the robot's compliant tube. Air enters the ring through its air inlet and is distributed throughout the ring, inflating it to compress the compliant tube and generate net fluid flow [3].

- Syringe:** provides and removes air from the ring through a tube, controlling its inflation and deflation
- Rigid Enclosure:** constrains outward expansion, forcing the ring to inflate inward against the compliant tube
- Air Inlet:** a point on the silicone ring where the air tube is inserted, created by removing a screw after the mold solidifies
- Servo:** would control the frequency of the syringe's pumping



References:

- [1] G. Unger, S. Patel, B. Kozyak, J. R. Raney, and C. Sung, "Liebau pumping enables valveless soft swimmer robot," in International Symposium on Experimental Robotics (ISER), 2025.
- [2] T. B. Fredenburg, T. R. Johnson, and M. D. Cohen, "The Fontan procedure: Anatomy, complications, and manifestations of failure," RadioGraphics, vol. 31, no. 2, pp. 453–463, Mar. 2011, doi: 10.1148/rg.312105027.
- [3] J. Anatol, M. García-Díaz, C. Barrios-Collado, J. A. Moneo-Fernández, F. Castro-Ruiz, and J. Sierra-Pallares, "Experimental characterization of an asymmetric valveless pump based on soft robotics technology," Physics of Fluids, vol. 35, no. 6, Art. no. 061904, June 2023, doi: 10.1063/5.0150978.

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